



Construction of kill webs with heterogeneous UAV swarms in dynamic contested environments

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Abstract

With the concept of "mosaic warfare," a novel combat style that involves constructing "kill webs" with unmanned aerial vehicle (UAV) swarms has emerged. However, little research has focused on this specific task scenario, particularly concerning the self-organization and adaptive collaboration of heterogeneous combat units in dynamic contested environments. Considering the scales and highly dynamic natures of such swarms, an adaptive communication network mechanism is developed based on the Molloy-Reed criterion. In contrast with common offline/noncombat task scenarios, the self-organization process is refined through agent-based modeling, and a combat effectiveness evaluation is introduced to provide enhanced task execution incentives. The proposed dynamic consensus-based coalition algorithm (DCBCA) addresses UAV intelligence defects such as "confusion," "forgetfulness," and "recklessness" during the dynamic target selection process, enabling effective bottom-up kill webs construction. Extensive simulation results demonstrate that the algorithmic system outlined in this paper can support the efficient and resilient operations of large-scale heterogeneous UAV swarms. The DCBCA outperforms the dynamically improved consensus-based grouping algorithm (CBGA) and the consensus-based timetable algorithm (CBTA) in terms of performance and convergence speed.

Keywords Heterogeneous unmanned aerial vehicle swarm · Kill webs construction · Complex adaptive systems · Dynamic adaptive networks · Resilience

Introduction

Due to the complexity of missions and the load limitations of single units, the UAV swarm combat style has been developed. Through the self-organization and adaptive behaviors of myriads of individual platforms, UAV swarms can conduct diverse combat missions [1–4]. During the mission execution

process, multiple combat platforms collaborate to construct cross-regional kill webs [5] similarly to the distributed, self-organizing group behaviors of biological communities [4, 6].

The "mosaic warfare" concept that has arisen in recent years is highly compatible with the aforementioned mission scenario [7]. In this paradigm, networked heterogeneous combat platforms collaborate to unleash desired combat effectiveness at defined times and locations, and the absence of specific components does not impede system performance. Thus, a large-scale heterogeneous UAV swarm combat system for kill web construction should possess the following properties.

- A fully distributed architecture for system flexibility and resilience.
- An adaptive communication network that can support stable and agile collaboration among individual UAVs.
- A self-organized driving scheme tailored to high-dynamic combat scenarios.
- Bottom-up information decisions for kill web construction.

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A robust communication network is essential for real-time decision-making in a UAV swarm. However, recent studies have primarily focused on predetermined mission scenarios [1] with relatively small system scales [6, 8]. Under such settings, a fully connected static communication network is assumed by default. In light of the large scales, payload constraints [9], and high dynamics [10] of UAV swarms in practical combat scenarios, this assumption must be revised. While some adaptive communication network models [11, 12] have been proposed for UAV swarms, they are mostly limited to reconnaissance tasks [2] and exhibit only flat network topology fluctuations. Thus, these models are not yet suitable for the construction of kill webs, which involve frequent cross-regional collaboration among heterogeneous UAVs, resulting in drastic network structure variations.

With multidimensional interweaving properties (communication-physical-mission [13]), a UAV swarm can be categorized as a complex adaptive system (CAS). By utilizing its advantages in terms of portraying the heterogeneity, nonlinearity, and emergence of CASs, agent-based modeling (ABM) has been widely applied to design self-organization behaviors that drive combat systems [3, 4, 7–9, 14, 15]. By setting several simple behavioral rules for individual agents (this term is interchangeable with UAVs in this paper unless it would cause confusion), a system eventually achieves complex bottom-up interactions. In practical applications, these rules are usually translated into a mechanical model [8, 16]. However, to the best of the authors' knowledge, no such model is currently tailored for kill web construction.

Apart from behavior driving, task "incentives" are also the keys to propelling the effective self-organization of combat systems. In common offline/noncombat scenarios [17, 18], the metrics used to quantify incentives are primarily confined to distance and time consumption levels. Although certain mission orientations have been incorporated to augment these metrics, they are mostly geared toward reconnaissance missions with homogeneous units [2, 11, 12], thus failing to capture the nonlinear combat effectiveness of networked heterogeneous UAVs.

The construction of kill webs with UAV swarms can be abstracted as a dynamic task assignment problem involving multiple agents [7]. For decentralized multiagent systems (MASs) such as UAV swarms, market-based methods are widely employed [19–21], where the task manager/auctioneer plays a pivotal role in the task announcement, bid evaluation, and task delegation steps. By further introducing a message exchange mechanism, the task manager/auctioneer can be removed, leading to a wholly distributed negotiation process.

After an extensive literature investigation, the authors deduced that this kind of fully distributed algorithm, represented by the consensus-based bundling algorithm (CBBA)

[22], most closely resembles the bottom-up information decision process employed for kill web construction. However, it is oriented to homogeneous multiagent task assignment scenarios under static conditions.

To enhance its applicability in various scenarios, numerous modified versions of the CBBA have been proposed, focusing primarily on dynamic [23–28], heterogeneous collaboration [29–32], dynamic networking [25, 27, 28, 33], and contested [33] aspects. Notably, in these works, dynamic network changes only accounted for agent communication limits; a fully connected network was still assumed within the communication range of a single agent. To the best of the authors' knowledge, no study has comprehensively considered the expansion of kill web construction task scenarios encompassing the four aspects described above.

The dynamic and heterogeneous nature of collaboration increases the complexity of the kill web construction task, leading many studies to improve the convergence speeds and reduce the communication costs of their algorithms (these two aspects are often intertwined). To adapt to dynamic scenarios, relevant studies [24, 26] have attempted to strike a balance between solving speed and solution quality, aiming to reduce the sensitivity of their systems to environmental uncertainties. Reference [34] indicated that this is an effective measure for enhancing the efficiency of task execution in large-scale distributed systems, which can be referred to as the "insensitivity principle."

To assign a single task to multiple collaborative agents, the consensus-based group algorithm (CBGA) [30] was proposed. Furthermore, temporal considerations are embedded a priori in the consensus-based timetable algorithm (CBTA) [32] to achieve synchronized collaboration. The insensitivity principle has been widely applied to address the algorithmic inefficiency induced by heterogeneous collaboration. In Ref. [32], an agent could only abandon the same task a limited number of times, and in Ref. [29], agents were allowed to finalize their decisions after some fixed waiting time, even before algorithmic convergence was achieved.

Although the above measures can reduce the negative impact caused by the uncertainty of dynamic "pop-up" targets and the task switches of cooperative agents, they are all algorithmic structure modification techniques. Intuitively, geographical criteria [35], i.e., the distances between agents and tasks in specific mission scenarios, can also be used to design the insensitivity principle, but work in this area is still rare. Furthermore, the optimal values of the critical parameters in the insensitivity principle have not been studied in conjunction with specific task scenarios; instead, they are arbitrarily assigned. Moreover, considering the excessive convergence steps [29, 32] and the communication bandwidth limits [36] required under dynamic conditions, it is unrealistic for large-scale systems to wait until the task assignment process is complete before executing their tasks.

The main contributions in this paper toward bridging the gaps clarified above are as follows.

1. A dynamic communication network model for adapting UAV swarms to stark battlefield environments is proposed.
2. Different self-organizational behavior methods are categorized for UAV swarms and oriented toward constructing kill webs.
3. Nonlinear combat effectiveness is incorporated into the task incentives of individual UAVs.
4. The CBBA is tailored for applications in dynamic, heterogeneous collaboration, dynamic networking, and contested scenarios.
5. The insensitivity principle is designed with geographical criteria to achieve improved algorithmic convergence.

The rest of the paper is organized as follows. Sect. "Dynamic communication network design for the UAV swarm" introduces the dynamic communication network design. An agent-based complex adaptive system model is established in Sect. "Agent-based complex adaptive system model". The dynamic consensus-based coalition algorithm (DCBCA) is proposed in Sect. "Kill webs construction based on the DCBCA". The results of a conducted simulation study are discussed in Sect. "Simulation results and analysis". This paper is concluded in Sect. "Conclusions and future work", and future work prospects are offered.

Dynamic communication network design for the UAV swarm

This section aims to design a dynamic and adaptive communication network for UAV swarms by considering the communication range limitations and the highly dynamic nature during missions.

Initial communication network

Mapping UAV swarms into communication networks, where the nodes denote UAV individuals and links represent the connections between two UAVs. According to [12], a static UAV swarm communication network is first constructed under communication-constrained conditions. The size of the initial network G_0 is set to be m_0 , with complete connections among the m_0 nodes, and then the new node i adds a new link to any node j in the existing network with the following probability.

$$P_{i \rightarrow j}(i, G_t) = \begin{cases} \frac{(k_j + \varepsilon)F(d_{i \rightarrow j})}{\sum_{l=1}^{N_t} F(d_{i \rightarrow l})(k_l + \varepsilon)}, & \sum_{l=1}^{N_t} F(d_{i \rightarrow l}) > 0, \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where k_j denotes the degree of freedom of node j , N_t is the number of nodes in the current network, G_t is the existing network, ε is the adjusting factor, and the distance effect function $F(d_{i \rightarrow j})$ is given as follows:

$$F(d_{i \rightarrow j}) = \begin{cases} 1, & d_{i \rightarrow j} < \eta r_c \\ \frac{(r_c - d_{i \rightarrow j})}{(1 - \eta)r_c}, & \eta r_c \leq d_{i \rightarrow j} < r_c, \\ 0, & d_{i \rightarrow j} \geq r_c \end{cases} \quad (2)$$

where r_c is the range of communication, $d_{i \rightarrow j}$ is the distance between nodes i and j , and η is the distance influential factor.

The generation process of the initial static communication network is illustrated in Fig. 1, some rules that need special mention are as follows:

- The order in which nodes join the network is preassigned randomly.
- If multiple nodes are connectable in the communication range of node i , it randomly selects one based on Eq. (1).
- If there are no connectable nodes currently, node i waits for the network to grow before attempting a connection.

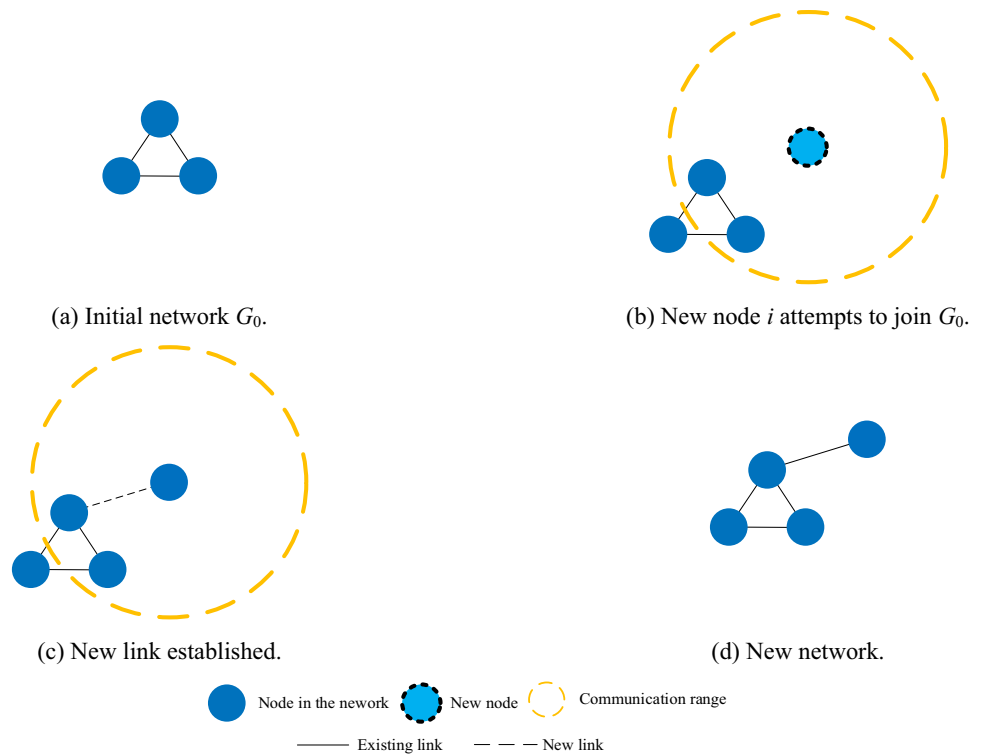
Finally, a static communication network $G = \{n_i\}$ of size N is obtained, originating from an initial size of m_0 . The key elements of the network can be summarized as:

- The node information $n_i = \{x_i, y_i, k_i\}$, where (x_i, y_i) represents the coordinates of node i .
- The connected cluster $C_i = \{n_{i,j}^C\}$, where $n_{i,j}^C$ denotes the node connected to node i .
- The neighbor list $NL_i = \{n_{i,j}^{NL}\}$, where $n_{i,j}^{NL}$ is the neighbor node of node i in its communication range.

Dynamic adaptive mechanisms

Considering the dynamic evolution of the swarm network in harsh battlefield environments, Algorithm 1 is used to achieve dynamic adaptive adjustment of the communication links between nodes.

The neighbor list NL_i of each node i undergoes constant changes during the self-organized movement. Accordingly, the distance between node i and any node $n_{i,j}^C$ in its connected cluster C_i may exceed the communication range r_c . In such cases, node $n_{i,j}^C$ is identified as an invalid node and

Fig. 1 Initial communication network generation

removed from the connected cluster C_i (lines 3–4), the link between each other is severed, thus the network connectivity is compromised.

For network robustness, each node performs a dynamic adaptation of the network. In [14], the giant component is used as an indicator of the network's structural integrity, with its presence determined through the MOLLY-REED criterion [37]. Upon identifying the absence of this component, each node will establish additional connections according to Eqs. (1) and (2). Cohen and Erez [38] derived the equivalent mathematical formulation of the MOLLY-REED criterion as

$$\kappa \equiv \frac{\langle k^2 \rangle}{\langle k \rangle} > 2. \quad (3)$$

where $\langle k \rangle$ denotes the average degree of freedom of nodes in the network. Calculation of the MOLLY-REED criterion, as described in Eq. (3), necessitates comprehensive information concerning the degree distribution of all network nodes. To simplify this determination, nodes may employ the average consensus technique [36] to approximate the values of $\langle k \rangle$ and $\langle k^2 \rangle$ based on the local information from connected nodes within a single hop, as shown in Eq. (4).

$$\langle k^* \rangle_i = \frac{\sum n_j^{C_i} k_j^{C_i}}{|C_i|}, \quad |C_i| > 0. \quad (4)$$

where $\langle k^* \rangle_i$ denotes the estimated value of $\langle k \rangle$ or $\langle k^2 \rangle$ by node i and $|C_i|$ is the scale of the connected cluster C_i .

Algorithm 1 Dynamic adaptive network

```

1. for each node  $i \in G$  do // Remove invalid nodes from the connected cluster  $C_i$ 
2.   Update  $NL_i$ 
3.   if  $n_{i,j}^C \in C_i$  and  $n_{i,j}^C \notin NL_i$  then
4.     Remove  $n_{i,j}^C$  from  $C_i$  and update  $k_i, k_{n_{i,j}^C}$ 
5. for each node  $i \in G$  do // Add new links to the network  $G$ 
6.   if  $|NL_i - C_i| > 0$  and  $k_i < k_{\max}$  then
7.     Compute  $\kappa_i$  from Eq. (5)
8.     while  $\kappa_i \leq 2e^\delta$  and  $k_i < k_{\max}$  and  $|NL_i - C_i| > 0$  do
9.       Add one new link from Eq. (1)
10.      Update  $C_i, k_i, k_{n_{i,j}^C}$ 
11.      Compute  $\kappa_i$  from Eq. (5)
12. return  $G$ 

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In this study, considering the estimation error and heterogeneity, a certain margin is set for the critical threshold value of κ . At the same time, to avoid information blocking caused by unrestricted growth of "hub" nodes, the maximum limit of each node's degree of freedom is set to $k_{\max}$. To conclude,

the improved activation condition for node adaptation is

$$\begin{cases} \kappa_i = \frac{\sum k_{n_j}^2}{\sum k_{n_j}^{c_i}} \leq 2e^\delta, \\ k_i < k_{\max}, \end{cases} \quad (5)$$

where κ_i is the estimated value of κ by node i ; in particular, when $|C_i| = 0$, node i directly determines $\kappa_i = 0$. δ is the margin factor.

During the mission, node i establishes a new link to node $n_{i,j}^{NL}$ in the neighbor list as long as Eq. (5) is satisfied (lines 9–11). The adaption process ceases upon failing to fulfill the activation condition or achieving complete connectivity with all nodes in the neighbor list (line 8). The properties of the dynamic communication network can be summarized as follows.

- The communication range of each single node is constrained (Eqs. (1), (2)).
- The dynamic adaption process is bottom-up based on each node's local information (Eqs. (3), (4)).
- The criteria for assessing the network's structural integrity have been moderately enhanced (Eq. (5)).
- The maximum value of each node's degree of freedom is limited (Eq. (5)).

Agent-based complex adaptive system model

In this section, the autonomous combat process of a UAV swarm is divided into multiple segments, followed by a study of corresponding behavior-driven approaches and task incentive design.

Composing the combat process

Inspired by typical combat scenarios in [3, 4, 7, 9, 12, 14, 27], the multiple motion states and transfer rules of a UAV unit for kill webs construction are shown in Fig. 2. The key elements involved in this issue are as follows [28].

(1) List of UAVs.

Let $U = \{U_1, \dots, U_{N_S}, U_{N_S+1}, \dots, U_{N_S+N_A}, U_{N_S+N_A+1}, \dots, U_{N_S+N_A+N_J}, \dots, U_{N_U}\}$ denotes N_U UAVs in the heterogeneous UAV swarm. These are, in order, N_S ISR (Intelligence, Surveillance, and Reconnaissance)-type, N_A attacker-type, N_J jammer-type, and N_D decoy-type UAVs, and the type of UAV U_i is represented by $type(U_i) \in \{S, A, J, D\}$.

(2) List of targets.

Consider N_T preconfigured targets $T = \{T_1, \dots, T_{N_T}\}$ in the mission scenario, each with multiple simultaneous task requirements $T_j = \{T_j^{type}\}$. For instance, $T_j = \{T_j^S, T_j^A\}$ indicates the ISR and attack tasks for target T_j . The task requirements are quantified by the number of UAV types needed as $D(T_j) = \{n_j^{type}\}$. For example, $D(T_j) = \{n_j^S, n_j^A\}$ means that n_j^S ISR-type UAVs and n_j^A attacker-type UAVs are necessary for a joint strike mission against target T_j .

(3) UAV coalitions.

The combat coalition against target T_j is represented by $G(T_j) = \{U_i^{T_j}\}$. Furthermore, $G(T_j^{type}) = \{U_{i,j}^{type}\}$ represents a group of homogeneous UAVs against target T_j . For example, $G(T_j^A) = \{U_{1,j}^A, U_{2,j}^A\}$ indicates that two attacker-type UAVs are currently aiming at T_j .

(4) Target selection profile.

The traditional top-down task assignment is translated into a bottom-up target selection profile of individual UAVs as $\alpha = \{a_i, \dots, a_N\}$, and element a_i represents the target selected by UAV U_i . For example, if $U_i \in G(T_j)$, then $a_i = T_j$. In particular, if UAV U_i does not have any target selected, then $a_i = \emptyset$. Generally, the composition of α directly reflects the construction of kill webs.

The mission engagement states of UAV U_i is defined as $es_i \in \{0, 1, 2\}$, which sequentially indicate uncommitted, committed, and sticky states. Moreover, its motion state is defined as $ms_i \in \{-1, 0, 1, 2, 3\}$, which sequentially means destroyed, sneak-like move, trajectory flight, orbiting wait, and synchronized strike.

"Uncommitted" refers to a UAV performing a sneak-like move in the mission space without being assigned to a specific target.

"Committed" indicates that the UAV is on a flight trajectory to a target after being summoned, but may adjust its trajectory if a more valuable target is identified. If the UAV discovers that a more capable UAV has already been assigned to the target, it will be considered outbid and revert to the uncommitted state.

"Sticky" denotes that the UAV remains fixed on the selected target until reaching a designated task location. Once there, it enters the orbiting wait state to establish a temporary combat coalition with other UAVs. Once all coalition members are in place, a synchronized strike on the target is conducted. After that, the coalition dissolves, and each UAV returns to the uncommitted state.

If a target has low priority, a UAV in the orbiting wait state may remain there for a prolonged period without forming a complete coalition, becoming "stuck." To resolve this, the UAV must actively disengage and revert to the uncommitted

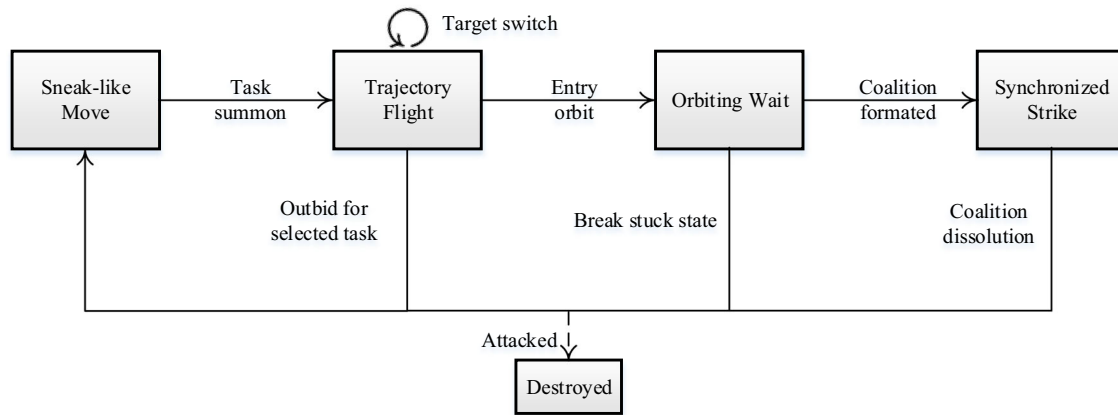


Fig. 2 Motion states and transfer rules of UAV units

state [30]. Furthermore, under contested conditions, the UAV may be attacked at any mission stage and destroyed.

Behavior-driven approaches

This subsection provides behavior-driven approaches for the diverse motion states of individual UAVs during mission execution.

(1) Sneak-like move.

Similar to [14, 16], an agent-based model is employed, each UAV experiences a series of forces influencing its stochastic behaviors. This paper delineates three principles that govern the self-organization of individual UAVs: collision avoidance, target attraction, and boundary bounce-back, corresponding to respective forces denoted as repulsion (r), attraction (a), and bounce (b) on the UAV U_i with its position $\mathbf{x}_i$ as

$$\begin{cases} F_i^r = \frac{\mathbf{x}_i - \bar{\mathbf{x}}_s}{\|\mathbf{x}_i - \bar{\mathbf{x}}_s\|^2 + \varepsilon_s}, \\ F_i^a = \bar{\mathbf{x}}_{i,t} - \mathbf{x}_i, \\ F_i^b = \frac{\Delta x}{\Delta x^2 + \varepsilon_b} + \frac{\Delta y}{\Delta y^2 + \varepsilon_b}, (\Delta x, \Delta y < 1) \end{cases} \quad (6)$$

where $\bar{\mathbf{x}}_s$ is the average position of the UAVs within the separation domain, $\|\cdot\|$ denotes the Euclidean norm, $\bar{\mathbf{x}}_{i,t}$ is the average position of targets recognized by U_i , Δx and Δy are the distances of the this UAV in the x and y directions to the boundary of the battlefield space, ε_s and ε_b are all adjusting factors.

Finally, the resultant force on UAV U_i is

$$\mathbf{F}_i(t) = \omega_r F_i^r + \omega_a F_i^a + \omega_b F_i^b, \quad (7)$$

where ω is the adjusting factor of each component force.

Assuming a constant mass of each UAV, an Euler integration method is used to update the position $\mathbf{x}_i$ and velocity $\mathbf{v}_i$ after each time step Δt .

$$\begin{cases} \mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t) + \mathbf{F}_i(t) \Delta t, \\ \mathbf{x}_i(t + \Delta t) = \mathbf{x}_i(t) + \mathbf{v}_i(t) \Delta t, \end{cases} \quad (8)$$

(2) Trajectory flight.

The coalition coordinated attack involves three key steps: approach, enclose, and synchronized strike [3]. Each UAV draws a circular path with equidistant points around the target for enclosing and waiting. The UAV heterogeneity creates gradient disparities of the circular paths, aiding in optimal attack formations. Different discrete points are used as end-points for trajectory planning and the one with the earliest expected arrival time is defined as the best entry point.

Considering the large scale of the swarm, a lightweight optimal control model for trajectory planning [27] is adopted, with the following input at any moment:

$$u_i(t) = \frac{4}{t_f^{i,j} - t} [\mathbf{v}_i(t_f^{i,j}) - \mathbf{v}_i(t)] + \frac{6}{(t_f^{i,j} - t)^2} [\mathbf{x}_f^{i,j} - \mathbf{x}_i(t) - \mathbf{v}_i(t_f^{i,j})(t_f^{i,j} - t)], \quad (9)$$

where $\mathbf{x}_f^{i,j}$ is the final discrete point position of the trajectory to target j , and $t_f^{i,j}$ is the estimated time to arrive at it. By tuning the value of $t_f^{i,j}$ to restrict the maximum flight speed v_m and the maximum angular speed ω_m , the entire flight trajectory is planned as discrete sequence points based on the time series.

(3) Orbiting wait.

In this phase, the UAV visits discrete points one by one to orbit the target. The motion model is simplified by maintaining the UAV's velocity tangent to the circular orbit at a constant speed.

Task incentive design

This subsection integrates combat effectiveness into task incentives for guiding UAV mission execution.

A "kill-web" comprising multiple "kill chains" is abstracted to represent the nonlinear combat effectiveness aggregation within the combat coalition. The combat effectiveness of a kill chain is described using operational cycle theory [39], where each UAV in the swarm is considered an independent combat node.

First, combat effectiveness is assigned to each type of combat node, namely, ISR effectiveness E_S , attack effectiveness E_A , jammer effectiveness E_J , and decoy effectiveness E_D . Second, a transfer rate of combat effectiveness between different combat nodes is determined, factoring in response speed. Based on the shortest communication topological distance d_{ik} between node i and node k , the response speed between nodes is represented as $\Delta^{d_{ik}}$, where $\Delta \in (0, 1]$ is the distance sensitivity factor.

The mission space targets are classified into two categories: I-type and II-type. I-type targets, such as airport runways, fuel depots, or maintenance plants, lack countermeasure capabilities. In contrast, II-type targets, primarily directed at enemy air defense positions, possess countermeasure capabilities.

A combat coalition oriented to I-type targets requires ISR and attack nodes to achieve the desired destructive impact, i.e., $T_j = \{T_j^S, T_j^A\}$. The nodal transfer matrix $\mathbf{M}$ representing the transmission of "kill chain" effectiveness between ISR-attack-target nodes is defined as

$$\begin{cases} \mathbf{M}_{TS} = (E_S)_{1 \times |G(T_j^S)|}, \\ \mathbf{M}_{SA} = (\Delta^{d_{ik}})_{|G(T_j^S)| \times |G(T_j^A)|}, \\ \mathbf{M}_{AT} = (E_A)_{|G(T_j^A)| \times 1}, \end{cases} \quad (10)$$

where $\mathbf{M}_{TS}$ for transmission between targets and ISR nodes, $\mathbf{M}_{SA}$ for transmission between ISR nodes and attack nodes, and $\mathbf{M}_{AT}$ for transmission between attack nodes and targets. Then, the nonlinear combat effectiveness of the combat coalition against target T_j is

$$E_j = \mathbf{M}_{TS} \mathbf{M}_{SA} \mathbf{M}_{AT}. \quad (11)$$

For example, a combat coalition targeting T_j consists of two ISR nodes and one attacker node, with their respective

combat effectiveness being E_S^1, E_S^2 , and E_A^1 . The topological distances between two ISR nodes and the attacker node are d_1 and d_2 . In that case, the transfer matrices are then $\mathbf{M}_{TS} = [E_S^1, E_S^2]$, $\mathbf{M}_{SA} = [\Delta^{d_1}, \Delta^{d_2}]^T$, and $\mathbf{M}_{AT} = [E_A^1]$.

For II-type targets, the attacker-type UAVs must complete the strike task under the cover of the jammer-type UAVs and the decoy-type UAVs, i.e., $T_j = \{T_j^S, T_j^A, T_j^J, T_j^D\}$. Taking the example of a jammer node k covering an attack node i , the covered attack effectiveness is

$$E_{A,J}^{\text{cover}}(i, k) = E_A E_J \frac{\Delta^{d_{ik}}}{\sum_{U_l \in G(T_j^A)} \Delta^{d_{lk}}}, \quad (12)$$

It can be understood that the jammer resources in a combat coalition are allocated by different attacker nodes in proportion to their response speed.

Then, the attack effectiveness of attack node i under the cover of all jammer nodes and decoy nodes in the coalition is

$$\begin{aligned} E_A^{\text{cover}}(i, G(T_j^J) + G(T_j^D)) \\ = \left(\sum_{U_k \in G(T_j^J)} E_{A,J}^{\text{cover}}(i, k) + \sum_{U_l \in G(T_j^D)} E_{A,D}^{\text{cover}}(i, l) \right) + E_A \sigma, \end{aligned} \quad (13)$$

where σ indicates the attack effectiveness under the condition without cover combat resources. The cover effectiveness of node i is defined as $E_{\text{cover}} = E_A^{\text{cover}} / E_A$. When $E_{\text{cover}} > 1$, E_{cover} is set to 1. Then, the transfer matrix between attack nodes and targets for II-type targets is

$$\mathbf{M}_{AT} = (E_A E_{\text{cover}})_{|G(T_j^A)| \times 1}. \quad (14)$$

The combat effectiveness of the current nonlinear kill webs formed by the UAV swarm over the whole mission area is

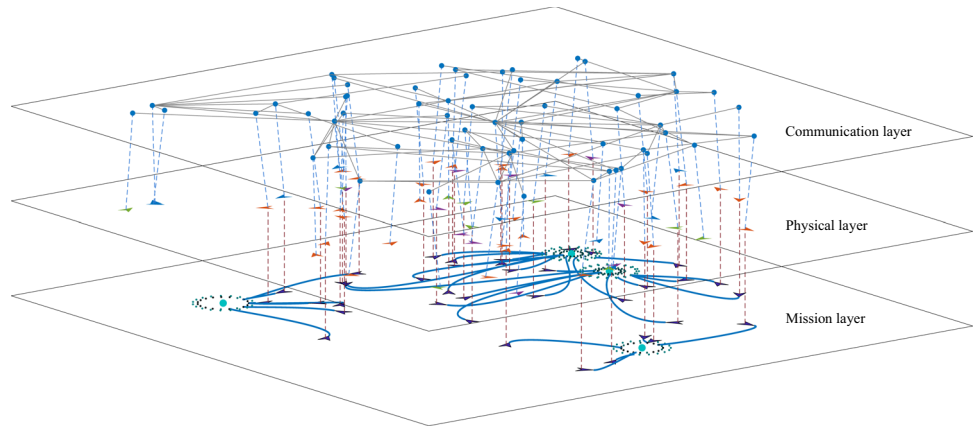
$$E_G = \sum_{j=1}^{N_T} E_j. \quad (15)$$

The expected mission rewards available to the combat coalitions at the current moment t are obtained by

$$r_{T_j}(G(T_j), t) = \bar{r}_{T_j} \exp\left(\frac{E_j}{DE_j} - 1 - \lambda \sum (ET A_j^{\text{type}} - t)\right), \quad (16)$$

where $\bar{r}_{T_j}$ is the nominal reward of target T_j , DE_j is the effectiveness corresponding to the target's demand for each type of UAV $D(T_j)$. If $E_j > DE_j$, $E_j = DE_j$ is set. λ is the

Fig. 3 Heterogeneous UAV swarm complex adaptive system



adjusting factor, and ETA_j^{type} represents the estimated time for each homogeneous grouping team $G(T_j^{type})$ to reach the designated task execution location, defined as the arrival time of the last UAV in the team. If there are no members in the grouping team of that type, it is set to a large value of Q . Properties of Eq. (16) can be summarized as:

- Nonlinear combat effectiveness is incorporated, which is rare in other studies.
- Mission rewards increase with the completeness of the coalition and proximity to the task execution location.

To conclude, the complex adaptive system model of a heterogeneous UAV swarm is established. The system can be divided into three levels: communication, physical, and mission, as illustrated in Fig. 3.

Kill webs construction based on the DCBCA

Based on the established complex adaptive system model, this section focuses on further enhancing the CBBA algorithm [22], and the DCBCA algorithm is introduced to better adapt to the dynamic, contested mission scenario of constructing kill webs with a large-scale heterogeneous UAV swarm.

Problem description and modeling

Considering the dynamic nature of the battlefield, the combat process is divided into discrete time steps. After each time step Δt , UAV U_i communicates with neighboring UAVs in the connected cluster C_i and updates the physical state and local information. Based on the latest decision information, each UAV decides target selection by joining different combat coalitions, enabling the swarm to construct dynamic kill webs across targets. Taking UAVs U_i and U_k connected through direct communication as an example, the

self-organization and self-adaptation of each UAV in the construction process of kill webs are illustrated in Fig. 4.

Based on the description above, the construction of kill webs is abstracted as a distributed, bottom-up, dynamic multi-agent task assignment problem, and correspondingly, it can be modeled as the following optimization problem.

$$\max \quad \Re(\alpha, t) = \sum_{j=1}^{N_T} r_{T_j}(G(T_j), t) \quad (17)$$

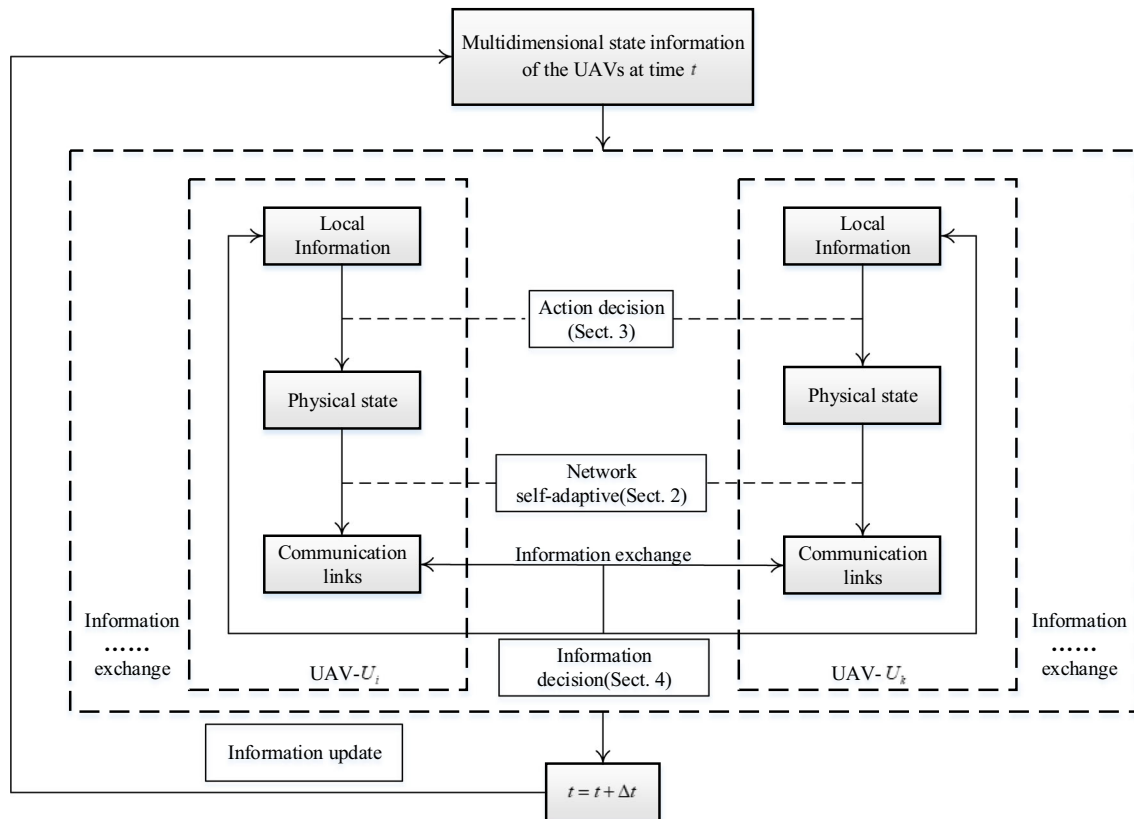
subject to

$$\begin{aligned} \sum_{j=1}^{N_T} \mathbb{I}(a_i = T_j) &\leq 1, \forall U_i \in U \\ \sum_{i=1}^{N_U} \mathbb{I}(a_i = T_j) &= n_j^k, \forall T_j \in T, type(U_i) = k \\ \sum_{i=1}^{N_U} \mathbb{I}(a_i = T_j) &= \sum_{D(T_j)} n_j^{type}, \forall (U_i, T_j) \in U \times T \end{aligned}$$

Equation (17) represents the current global mission rewards of the UAV swarm, where α is the target selection profile of individual UAVs. $\mathbb{I}(\cdot)$ is the indicator function that is unity if the argument is true and zero otherwise.

This paper enhances the optimization objective through a potential game strategy [11, 27], i.e. a decentralized way, to adapt to highly dynamic and uncertain combat environments. Individual UAVs are incentivized to maximize their immediate gain in a greedy manner and, in turn, drives the improvement of global mission rewards. The reward received by UAV U_i for performing a collaborative strike mission against target T_j by joining coalition $G(T_j)$ is defined as

$$\begin{aligned} r_i(T_j) &= \Re((T_j, \alpha_{-i}), t) - \Re((\emptyset, \alpha_{-i}), t) \\ &= r_{T_j}(G_{U_i}(T_j), t) - r_{T_j}(G_{-U_i}(T_j), t), \end{aligned} \quad (18)$$



The size of Y^i matches that of Z^i . In Y^i , y_{kj}^i represents UAV U_k 's bid for target $T_{i,j}$, as known by UAV U_i , all initially set to negative infinity. Per Eq. (16), smaller ETA_j^{type} values for same-sized coalitions imply higher mission rewards. For simplicity, UAV U_i 's bid for target $T_{i,j}$ is $-t_f^{i,j}$, reflecting the time cost to reach the task location.

(5) Target choice $\mathbf{c}_i$.

$\mathbf{c}_i = \{T_{i,j}\}$ indicates that UAV U_i currently selects target $T_{i,j}$. Otherwise, $\mathbf{c}_i = \emptyset$.

(6) Communication timestamp vector S^i .

S^i , sized $1 \times N_U$, records the latest moment when UAV U_i receives a message from its neighboring UAV.

DCBCA process

Algorithm 2 outlines the DCBCA structure, involving two phases: target selection and consensus. Each UAV selects the highest-reward target based on current situation information (lines 2–3). Since this process is developed from a distributed perspective based on different UAVs in a dynamic environment, conflicts in decisions can arise, leading to collaborative conflict resolution among neighboring UAVs (lines 4–5).

Algorithm 2 Dynamic consensus-based coalition algorithm

Initialization: $\tau_r = 0$ // The number of algorithm execution rounds in one time step

1. while $\tau_r \leq c_t$
2. for each UAV U_i s.t. $es_i \neq 2$ do // Phase 1: Target selection process
3. | Target selection () // Select best target $T_{i,j}$ and add it to $\mathbf{c}_i$
4. for each UAV U_i s.t. $ms_i \neq -1$ do // Phase 2: Consensus process
5. | Consensus () // Comm with $n_j^C \in C_i$, update $Z^i, Y^i, S^i, \mathbf{c}_i$ and es_i
6. $\tau_r = \tau_r + 1$
7. return $\alpha, Z^i, Y^i, S^i, \mathbf{c}_i, es_i, TL^i, DL^i$

Multiple iterations of the DCBCA process occur within each time step to determine the global target selection for the next step. The number of information interactions between neighboring UAVs per time step is set to c_t , matching the number of DCBCA rounds. Sect. "Simulation results and analysis" will provide a detailed analysis of the specific value of c_t , crucial for algorithm convergence and mission performance.

A. Target selection process.

Algorithm 3 Calculate mission rewards

1. $G(T_{i,j}) = \{U_k \mid z_{kj}^i = 1\}$
2. if $U_i \in G(T_{i,j})$ then
3. Remove U_i from $G(T_{i,j})$
4. Compute $r_{T_{i,j}}(G_{-U_i}(T_{i,j}))$ from Eq. (16)
5. if $y_{ij}^i = -\infty$ then
6. Compute $t_f^{i,j}$ to $T_{i,j}$ from Eq. (9)
8. Mark y_{ij}^i as $-t_f^{i,j}$
9. if $|G_{-U_i}(T_{i,j}^{type})| < n_{i,j}^{type}$ then
10. Compute $r_i(T_{i,j})$ from Eq. (18)
11. elseif $y_{ij}^i > y_{ij}^j$
12. Compute $r_i(T_{i,j})$ from Eq. (18)
13. elseif $y_{ij}^i < y_{ij}^j$
14. $r_i(T_{i,j}) = 0, ts_{i,j} = 0.5, BG_i(T_{i,j}) = G_{-U_i}(T_{i,j}^{type})$
15. return $TL^i, BG_i(T_{i,j}), r_i(T_{i,j})$

(1) Calculate mission rewards

During the target selection phase, if the UAV is currently in the sticky state, it directly skips this process and maintains the previous target choice $\mathbf{c}_i$. In other mission engagement states, calculating the mission reward $r_i(T_{i,j})$ for joining a combat coalition is the key procedure in selecting targets, as presented in Algorithm 3.

First, UAV U_i calculates the expected mission rewards to the combat coalition excluding itself, i.e., $\mathfrak{R}((\emptyset, \alpha_{-i}), t)$ (lines 2–4 of Algorithm 3). The calculation involves determining real-time communication connectivity within the coalition to evaluate combat effectiveness. However, under dynamic conditions, limited local perception poses challenges for UAVs in assessing this connectivity. Given that once all coalition members move to the orbiting wait state, they can execute a silent synchronized strike using basic local communication [3]. In such case, the "kill chain" transfer rate is disregarded with the distance sensitivity factor $\Delta = 1$. Consequently, the coalition's combat effectiveness E_j can be roughly estimated as

$$\tilde{E}_j = \begin{cases} \prod_{type \in \{S, A\}} E_{type} \cdot (|G(T_{i,j}^{type})| + \mu), & T_{i,j} \text{ is I-type} \\ \tilde{E}_{cover} \prod_{type \in \{S, A\}} E_{type} \cdot (|G(T_{i,j}^{type})| + \mu), & \text{otherwise} \end{cases} \quad (19)$$

$$\tilde{E}_{cover} = \sigma + \frac{\mathbb{I}(|G(T_{i,j}^A)| > 0)}{|G(T_{i,j}^{type})| + \mu}$$

$$\sum_{type \in \{J, D\}} E_{type} \cdot \left(\left| G(T_{i,j}^{type}) \right| + \mu \right), \quad (20)$$

where μ is the adjusting factor used to attract UAVs to initiate coalitions. Compared with Eqs. (10)–(13), one can summarize the key properties of the estimated coalition combat effectiveness:

- The calculation process remains unchanged, with the distance sensitivity factor Δ set to 1.
- It is only related to the composition of the coalition and not to the network structure.
- It can be calculated by the Winners matrix from the local perspective of individual UAVs.

Then, UAV U_i calculates $t_f^{i,j}$ using Eq. (9) to bid for target $T_{i,j}$ (lines 5–8 of Algorithm 3). If the coalition size hasn't met the requirement, it directly computes $\mathfrak{N}((T_{i,j}, \alpha_{-i}), t)$, thus obtaining the mission reward $r_i(T_{i,j})$ from Eq. (18). For saturated coalitions, UAV U_i compares $-t_f^{i,j}$ and $y_{lj}^i = \min \{y_{kj}^i \mid (\forall U_k \in G_{-U_i}(T_{i,j}^{type}))\}$. If $-t_f^{i,j} > y_{lj}^i$, indicating that UAV U_i joining the coalition is more advantageous, then UAV U_i calculates $r_i(T_{i,j})$ (lines 11–12 of Algorithm 3) and may substitute UAV U_l . Otherwise, $r_i(T_{i,j})$ is set to 0.

Algorithm 4 Target selection

```

1. Gain = 0 ;  $G(T_{i,j}^{type}) = \{U_k \mid z_{kj}^i = 1, type(U_k) = type(U_i)\}$ 
2. for each  $T_{i,j} \in TL^i$  do
3.   if  $ts_{i,j} = 0.5$  then
4.     if  $BG_i(T_{i,j}) = G(T_{i,j}^{type})$  then
5.       continue
6.     else
7.        $ts_{i,j} = 1$ ,  $BG_i(T_{i,j}) = \{ \}$ 
8.   if  $n_{i,j}^{type} > 0$  and  $ts_{i,j} = 1$  and  $|\mathbf{x}_i - \mathbf{x}_{i,j}| < \text{select limit}$  then
9.     Calculate gain  $r_i(T_{i,j})$  with Algorithm 3
10.    if  $r_i(T_{i,j}) > \text{Gain}$  then
11.      Gain =  $r_i(T_{i,j})$ ,  $\mathbf{c}_i = \{T_{i,j}\}$ 
12. if Gain > 0 then
13.   Update  $es_i$ ;  $z_{ij}^i$ ;  $y_{ij}^i$ ;  $G(T_{i,j}^{type})$ 
14.   if  $|G(T_{i,j}^{type})| > n_{i,j}^{type}$  then
15.      $z_{ij}^i = 0$ ,  $y_{ij}^i = -\infty$  //  $y_{ij}^i = \min \{y_{kj}^i \mid (U_k \in G(T_{i,j}^{type}), k \neq i)\}$ 
16. return  $es_i$ ,  $Z^i$ ,  $Y^i$ ,  $\mathbf{c}_i$ 

```

(2) Improvement of individual UAV intelligence

Forgetful defect

Each time re-entering the target selection process, avoiding repetitive and time-consuming reward calculations is

crucial for algorithm efficiency. This behavior can be seen as an intelligent defect of "forgetfulness," where UAVs forget their past failures in target selection.

UAV U_i can streamline this by recording the best grouping team $BG_i(T_{i,j})$ for target $T_{i,j}$. Team members are the same type as U_i , with bids exceeding y_{ij}^i . Target selection is triggered only if the team composition changes. Otherwise, the target status marker $ts_{i,j}$ is set to 0.5, bypassing the process. Refer to lines 13–14 of Algorithm 3 and lines 2–7 of Algorithm 4 for detailed steps.

Confused defect

During target selection, due to limited communications per time step, individual UAVs may struggle with optimal target selection, leading to frequent switches. This behavior is especially notable during the initial phase of dynamic pop-up target injection.

To expedite DCBCA convergence, addressing this confusion is crucial. This study proposes setting UAVs to a sticky engagement state ($es_i = 2$) once they reach a specific range from the target, termed the sticky limit. Notably, superior UAVs may exclude sticky UAVs from the coalition via a consensus process outlined in Algorithm 6. Like parameter c_t , the sticky limit significantly influences algorithm performance. Further in-depth research will explore this after introducing the "operational resource density" concept.

Reckless defect

Choosing distant targets hinders efficient "kill web" closure; UAVs selecting these targets face a higher risk of exclusion from the coalition by peers. Such behavior reflects the 'reckless' nature of UAVs in target selection. This paper introduces a constraint limiting UAVs to select targets within a defined range (select limit) to mitigate this issue.

The complete target selection process is detailed in Algorithm 4, with summarized properties as follows.

- By addressing three types of intelligent defects, the algorithm computation is reduced and convergence is accelerated.
- The insensitivity principle is once again employed by sticky limit and select limit to enhance the algorithmic efficiency.
- Beyond structural algorithmic modifications, geographical criteria are embedded within the insensitivity principle.

B. Consensus process.

In the consensus phase, UAVs exchange the battlefield and decision information with neighbors to achieve consensus on the situation and a conflict-free target selection scheme.

(1) Situation information sharing

Prior to conflict resolution, UAVs must achieve a uniform level of situation awareness by exchanging the target list TL_i

and the destroyed UAV list DL_i . Algorithm 5 illustrates the process using the example of UAV U_i receiving messages from UAV U_k . Lines 2–5 indicate that UAV U_i is informed of a new target by UAV U_k . In lines 6–9, UAV U_i learns by UAV U_k that target $T_{i,j}$ has been destroyed. In lines 10–12, UAV U_i receives news from UAV U_k that UAV $U_{k,m}$ has been destroyed.

In addition, UAVs must assess the timeliness of the received information to avoid outdated or invalid data that could disrupt decision-making. To address this, a communication timestamp vector S^i is designed, where element s_{ik} denotes the "latest moment" when UAV U_i receives a message from UAV U_k [22]. In this paper, the "latest moment" is defined as the cumulative number of communications between two UAVs, with both updating this value after each interaction. The larger the value of s_{ik} , the newer the message, and the update rule is

$$s_{ik} = \begin{cases} \sum \Delta t / \Delta t \times c_t + \tau_r, & g_{ik} = 1 \\ \max_{l: g_{il}=1} s_{il}, & \text{otherwise} \end{cases}, \quad (21)$$

where c_t is the number of information interactions between two neighboring UAVs per time step, τ_r is the cumulative number of communications within the current time step, $g_{ik} = 1$ represents that UAV U_i and UAV U_k are directly connected. For example, with Δt as 2 s and c_t

as 3, if UAV U_i and UAV U_k are connected directly, then after the first communication within the fourth time step, $s_{ik} = [(4 - 1) \times 2] / 2 \times 3 + 1 = 10$.

Algorithm 5 Situation info sharing

```

1. for each  $T_{k,j} \in TL^k$  do
2.   if  $T_{k,j} \notin TL^i$  then // Judging by coordinates  $\mathbf{x}_{k,j}$  and  $\mathbf{x}_{i,j}$ 
3.     Add  $T_{k,j}$  to  $TL^i$ ,  $Z^i = [Z^i, \mathbf{0}_{N_{i,j} \times 1}]$ ,  $Y^i = [Y^i, -\infty \cdot \mathbf{1}_{N_{i,j} \times 1}]$ 
4.     if  $es_i = 2$  then
5.       Mark  $es_i$  as 0
6.   elseif  $T_{k,j} = T_{i,j}$  and  $ts_{k,j} \leq 0$  and  $ts_{i,j} > 0$ 
7.     Mark  $ts_{i,j}$  as  $ts_{k,j}$ 
8.     if  $c_i = T_{i,j}$  then
9.       Mark  $es_i$  as 0
10. for each  $U_{k,m} \in DL^k$ 
11.   if  $U_{k,m} \notin DL^i$  then // Judging by the exclusive ID of  $U_{k,m}$  and  $U_{i,m}$ 
12.     //The adjusted row is row  $ID(U_{k,m})$ 
13.     Add  $U_{k,m}$  to  $DL^i$ ,  $Z^i = [\dots, \mathbf{0}_{N_{i,j} \times 1}, \dots]^T$ ,  $Y^i = [\dots, -\infty \cdot \mathbf{1}_{N_{i,j} \times 1}, \dots]^T$ 
13. return  $TL^i, DL^i, Z^i, Y^i, c_i, es_i$ 

```

(2) Conflict resolution

Based on the same situation awareness level, UAVs continue to exchange information for conflict resolution of the target selection scheme.

Algorithm 6 Consensus

```

//  $U_i$  receive the old info sent from  $U_k$  and generate new info
1. for each  $T_{k,j} \in TL^k$  do
2.   Share situation info with Algorithm 5
3.   for each  $type \in \{S, A, J, D\}$  do
4.     if  $n_{k,j}^{type} = 1$  then
5.       Consensus from CBBA
6.     else
7.       for each  $U_m \in G(T_{i,j}^{type})$  do // Updating the outdated info
8.         if  $(s_{km} > s_{im} \text{ and } m \neq i) \text{ or } m = k$  then
9.            $z_{mj}^i = z_{mj}^k, y_{mj}^i = y_{mj}^k$ 
10.        for each  $U_m \in G(T_{k,j}^{type})$  do // Adding the latest info
11.          if  $(s_{km} > s_{im} \text{ or } m = k) \text{ and } m \neq i \text{ and } U_m \notin G(T_{i,j}^{type})$  then
12.            if  $|G(T_{i,j}^{type})| < n_{k,j}^{type}$  then
13.               $z_{mj}^i = z_{mj}^k, y_{mj}^i = y_{mj}^k$ 
14.            else
15.              if  $y_{mj}^k > y_{mj}^i = \min\{y_{nj}^i\} (\forall U_n \in G(T_{i,j}^{type}))$  then
16.                if  $l = i$  then
17.                  Mark  $es_i$  as 0, clear  $c_i$ 
18.                   $z_{ij}^i = 0, z_{mj}^i = z_{mj}^k, y_{ij}^i = -\infty, y_{mj}^i = y_{mj}^k$ 
19.  $s_{ik} = \sum \Delta t / \Delta t \times c_i + \tau_r$ ;
20. return  $TL^i, DL^i, Z^i, Y^i, S^i, c_i, es_i$ 

```

In most CBBA algorithm modifications [22, 28, 30, 35], two neighboring UAVs must continuously exchange information until the algorithm converges before decision-making. For kill webs construction scenarios with the complexity of heterogeneous collaboration and dynamic uncertainty, achieving global convergence within one time step is impractical. Numerous numerical simulations suggest that for most physically meaningful scenarios, many assignments will be locked in the early stage of CBBA [29]. Therefore, inspired by Ref. [29, 32], in this paper, the value of c_t is set relatively small, but within a limited number of time steps, the swarm can also achieve conflict deconfliction.

When resolving conflicts in combat coalition formation $G(T_j)$, deconflicting each type of grouping team $G(T_j^{type})$ is essential (Algorithm 6, line 3). Conflict resolution for $G(T_j^{type})$ mirrors that in the CBBA algorithm [22] when the target demand for a specific UAV type n_j^{type} is 1 (lines 4–5). If $n_j^{type} > 1$, the conflict resolution process can be divided into two steps: updating the outdated information (lines 7–9) and adding the latest information (lines 10–18).

- In the case of third-party UAV U_m selecting target $T_{k,j}$, if UAV U_i perceives that the information it has is outdated compared to that from UAV U_k , it updates this information with the data from UAV U_k .
- Adding the latest information suggests that UAV U_i is unaware of UAV U_m 's selection of target $T_{k,j}$. Subsequently, it integrates the local information about $G(T_{i,j}^{type})$ with pertinent data from U_k to determine if U_m is aligned with the coalition against target $T_{k,j}$.

Properties of the consensus process can be highlighted as:

- It allows UAVs to make decisions before the algorithm reaches full convergence.
- It can resolve conflicts in the global target selection scheme caused by limited communication.
- It can partially compensate for the loss of optimality in assignments due to sticky UAVs.

Simulation results and analysis

Mission scenario and parameter setting

The battlefield environment is set as a $10 \text{ km} \times 10 \text{ km}$ square area ($L = 10 \text{ km}$). A heterogeneous UAV swarm with an initial size of N_U is randomly and uniformly deployed in the mission space. The ratio of each type of UAV is $N_S:N_A:N_J:N_D = 1:3:1:1$. The upper limit of the degree of freedom of a

single node is $k_{\max} = 15$. There are N_T targets in the mission scenario, where the number of I-type targets is the same as the number of II-type targets. Without specifying, the number of communication exchanges between neighboring UAVs per unit time step during mission execution is set as $c_t = 5$. The sticky limit = 5 km and select limit = 7 km are set.

Analysis of the network topology adaptive effect

To verify the superiority of the proposed communication network adaptive mechanism, it is compared with the method discussed by Bai et al. [12]. In this experiment, the size of the UAV swarm is set as $N_U = 60$, the number of targets is $N_T = 4$, and the range of communication is $r_c = 3 \text{ km}$. The network topology at the 5-time step is shown in Fig. 5.

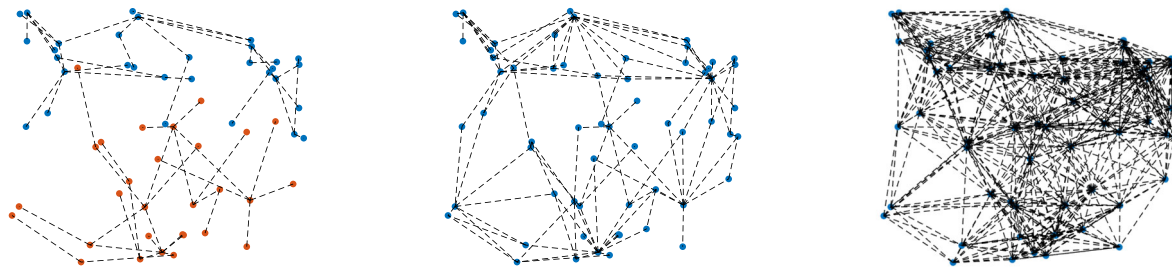
The communication network formed with the method proposed by Bai et al. [12] breaks into two major parts distinguished by red and blue markings. By contrast, based on the algorithm proposed in this paper, each node adds slightly more communication links so that the communication network is effectively linked as a whole, exhibiting strong robustness. If any node in the communication network is connected to all the neighboring nodes in its communication range, it is called the "fully connected" mode of the communication network. As shown in Fig. 5c, the number of communication links will surge in this mode.

To further investigate the impact of communication network adaptive mechanisms on combat effectiveness, the concept of "potential combat effectiveness" for heterogeneous UAV swarms is introduced. Considering a heterogeneous UAV swarm as a global kill webs capable of dispatching combat resources, the "potential combat effectiveness" P is expressed as

$$\begin{cases} P = \mathbf{M}_{TS} \mathbf{M}_{SA} (\mathbf{M}_{AT}^1 + \mathbf{M}_{AT}^2) / (2\varphi), \\ \mathbf{M}_{TS} = (E_S)_{1 \times N_S}, \\ \mathbf{M}_{SA} = (\Delta^{d_{ik}})_{N_S \times N_A}, \\ \mathbf{M}_{AT}^1 = (E_A)_{N_A \times 1}, \\ \mathbf{M}_{AT}^2 = (E_A E_{\text{cover}})_{N_A \times 1}, \end{cases} \quad (22)$$

where $(\mathbf{M}_{AT}^1 + \mathbf{M}_{AT}^2) / (2\varphi)$ indicates an equal number of targets of both types in the mission space and φ is a discount factor. For a fixed communication network topology, the information transmission efficiency is related to the number of communications between nodes per unit time step. Thus, the distance sensitivity factor based on the sigmoid function is defined as

$$\Delta \triangleq 2 \times \left(\frac{1}{1 + e^{-c_t}} - \frac{1}{2} \right), \quad (23)$$



(a) Method proposed by Bai et al. [19] (b) Method proposed in this paper. (c) Full-connected mode in [6, 27, 28].

Fig. 5 Communication network of the UAV swarm

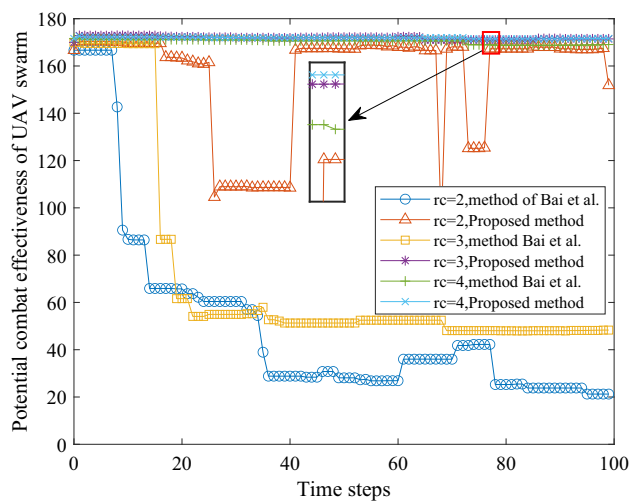


Fig. 6 Comparisons of the potential combat effectiveness under different communication conditions

Then, the range of communication r_c between UAVs is adjusted to examine the effect of two different adaptive mechanisms on the "potential combat effectiveness" of the UAV swarm, as shown in Fig. 6.

When the communication range is much limited as $r_c = 2$ km, both of these adaptive mechanisms can cause the communication network to become partially disconnected, resulting in a significant decrease in the potential combat effectiveness of the swarm. However, our proposed method enables the network to be restored more quickly after disconnection. This disparity becomes even more pronounced when $r_c = 3$ km. As $r_c = 4$ km, the communication network can maintain stable connectivity with both adaptive mechanisms. Nevertheless, in terms of potential combat effectiveness, our adaptive mechanism demonstrates superior performance.

Construction of kill webs with heterogeneous UAV swarms

By shifting the focus from the communication layer to the physical and mission layers, this section examines the construction of kill webs using a heterogeneous UAV swarm.

Without loss of generality, four different target parameter templates are designed for the mission scenario, two each for I-type targets and II-type targets. The firepower demand, optimal attack positions with gradient differences, and nominal rewards for these targets are shown in Table 1. The order of the values is ISR-attacker-jammer-decoy. The firepower demand is characterized by the target's requirement for the number of each type of UAV, and the radius of the circular path is used to indicate the optimal attack position.

First, a mission scenario with UAV swarm size $N_U = 60$ and number of targets $N_T = 4$ is used as an example to illustrate the process of constructing kill webs with a heterogeneous UAV swarm. At the initial moment, each target takes one of the target parameter templates from Table 1. When a target is destroyed, it is supplemented by a pop-up target, keeping the number of targets in the mission space unchanged. The pop-up target remains the same type as the replaced target, and the parameter settings are chosen randomly from Table 1. By adopting the above method, the impact of the randomness of the target location and UAV swarm distribution on the experimental results can be reduced. To achieve a Monte Carlo simulation, the terminal time is extended to 300 time steps.

The battlefield situation at four distinct time steps during the operation is depicted in Fig. 7 and the communication network is omitted for simplicity. As illustrated in Fig. 7a, the UAVs temporarily appear "overprovisioned" for multiple targets. However, after four time steps, as depicted in Fig. 7b, the UAV swarm resolves conflicts in the global target selection scheme and completes the construction of kill webs for full target coverage. At the 40th time step, the UAV swarm destroys the first target (the red pentagram at the bottom

Table 1 Parameter template for the targets

No.	Type	Firepower demand	Optimal attack position (km)	Nominal reward
1	I-type	1–6–0–0	0.45–0.38–/–/	0.6
2	I-type	1–5–0–0	0.42–0.31–/–/	0.5
3	II-type	1–6–4–3	0.43–0.37–0.26–0.17	1.8
4	II-type	1–4–2–3	0.43–0.39–0.23–0.17	1.5

of Fig. 7c), and a random target of the same type is generated in the mission space. By the 90th time step, the UAV swarm has destroyed three targets and maintains kill webs coverage of all targets. Therefore, it is proved that based on the developed algorithm system, heterogeneous UAV swarms can effectively construct kill webs in dynamic scenarios.

This paper focuses on the adaptive combat capabilities of heterogeneous UAV swarms of a specific size in a battlefield environment with a designated number of targets. Thus, the concept of operational resource density ρ is introduced:

$$\rho = |N_U| : |N_T| : |L^2| \quad (24)$$

where $|L^2|$ denotes the area of the two-dimensional square battlefield space. To verify the adaptability of the algorithm in different scenarios, UAV swarms of various sizes (30, 60, 90, and 120 UAVs) and a battlefield environment with different numbers of targets (2, 4, 6, and 8) are established. The cumulative rewards generated from the destruction of targets by UAV swarms over 300 time steps are used as a performance indicator. To mitigate the risk of losing research value due to long-term global communication network disconnections when the UAV swarm is relatively small, the communication range is appropriately relaxed as $r_c = 5$ km.

Figure 8 illustrates a clear upward trend in cumulative rewards as the UAV swarm size increases. However, when the UAV swarm size is fixed and the number of targets increases, there is not necessarily a corresponding increase in cumulative rewards (refer to the red boxes in the figure for the maximum cumulative rewards corresponding to various number of targets). This disparity arises because, in environments with many targets, the UAV swarm may fail to achieve full kill webs coverage, leading to some UAVs entering a "stuck" state and impeding the combat process. Thus, in practical combat applications, the size of the UAV swarm and the selection of tactics should be determined based on the operational resource density.

Analysis of the improvement of individual UAV intelligence

This section focuses on evaluating the effectiveness of the DCBCA in mitigating the "forgetful" and "confused" intelligence defects of individual UAVs. This analysis is crucial for

realizing the objectives outlined in Sect. "Dynamic Consensus-Based Coalition Algorithm (DCBCA)", which include reducing algorithmic computation and improving algorithm convergence speed.

Improvement in the "forgetful" defect

In this experiment, the number of trajectory planning performed by the UAV swarm for reward calculation in the target selection phase is used as an indicator. The parameter settings in this experiment are consistent with those in the mission scenario in Sect. "Construction of kill webs with heterogeneous UAV swarms", where $N_U = 60$ and $N_T = 4$.

As shown in Fig. 9, after the improvement, for the majority of the time steps, trajectory planning is nearly absent, except for a spike when a new target pops up. The sporadic minor upticks occur when some of the UAVs reach the selection limit for a target for the first time. The above results show that the computational workload of the algorithm is effectively reduced.

Improvement of the "confused" defect

Essentially, the "confused" defect arises from the battlefield's uncertain situational information. A UAV in a "confused" state frequently switches target selection, further exacerbating battlefield uncertainty. As demonstrated in [34], reducing the information received by individual agents can enhance the operational efficiency of large-scale distributed systems. This mechanism aims to lessen the perceived uncertainty in the mission environment by reducing the information received by individual agents. Therefore, as previously stated in Sect. "DCBCA process", to replicate this effect, a "sticky" mission state and a limited number of information interactions c_t are employed. The remaining parameter settings are consistent with those in Sect. "Improvement in the "forgetful" defect".

Figure 10a illustrates that when the sticky limit is set to 1, the UAV swarm experiences a severe "confused" state, resulting in highly fluctuating kill webs combat effectiveness curves. This suggests that individual UAVs frequently change their target selection. When $c_t = 7$, a "stuck" state even occurs, as indicated by a sharp drop in the purple curve. The cumulative rewards graph in Fig. 10b emphasizes this

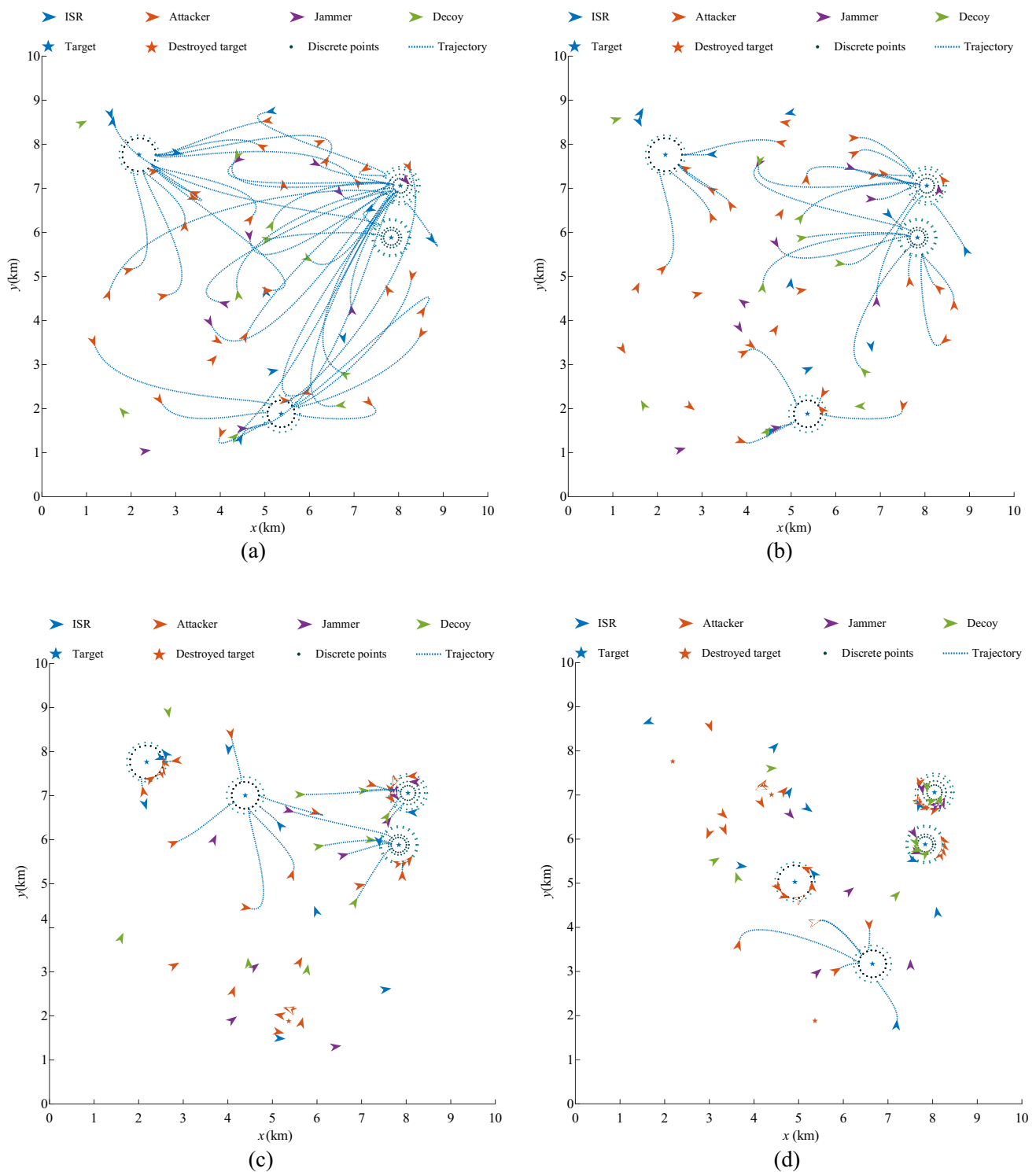


Fig. 7 Battlefield situation at different time steps: (a) 1, (b) 5, (c) 40, (d) 90

phenomenon. With $c_t = 7$, the cumulative rewards of the UAV swarm grow slowly. Conversely, with an appropriate c_t value (3 or 5), the swarm maintains an optimal level of situation awareness, prompting each UAV to make decisive target selections and expedite mission completion progress.

In contrast, the "confused" symptom of the swarm is better mitigated with sticky limit = 3, as depicted in Fig. 11a, where significant, periodic fluctuations are seldom observed. From the cumulative reward curve shown in Fig. 11b, there is still a contradiction between the mission execution efficiency and

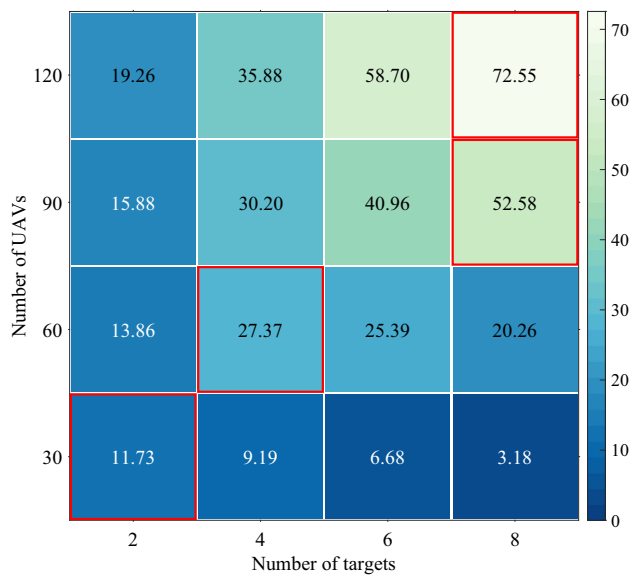
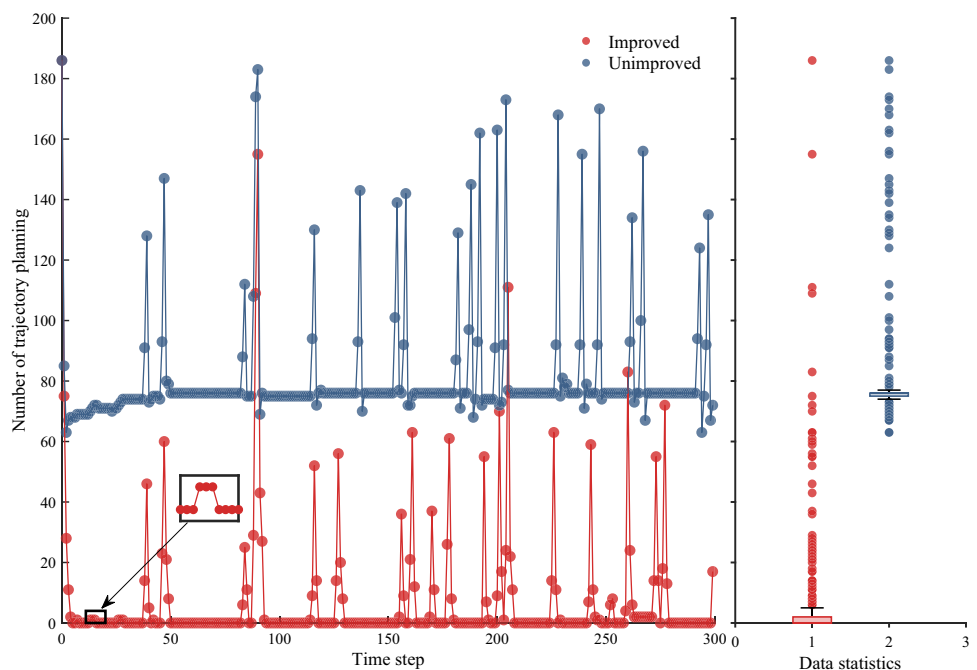


Fig. 8 Impact analysis of the effect of the operational resource density on the cumulative rewards

the situation awareness capability of the UAV swarm when larger values of c_t are considered.

The "confused" symptom of the swarm is further alleviated with a sticky limit of 5, as shown in Fig. 12a. Analysis of the cumulative reward curves reveals that when $c_t \leq 5$, there is a positive correlation between mission execution efficiency and the value of c_t , suggesting an enhanced ability of the swarm to process situational information. However, when $c_t = 7$, the swarm still suffers information overload, leading to a blockade in the mission execution.

Fig. 9 Improvement in the "forgetful" defect

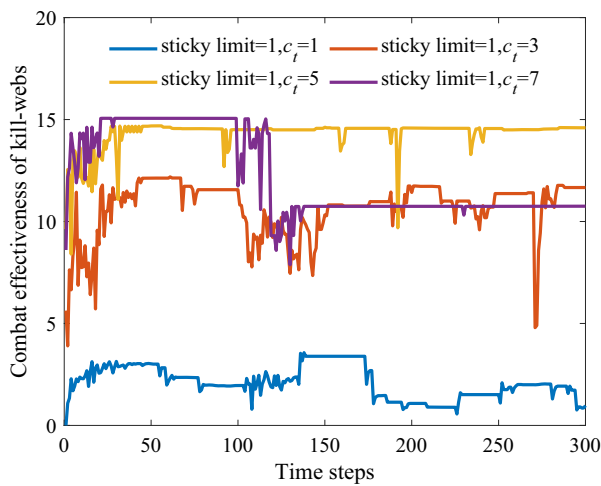


With sticky limit = 7, in terms of cumulative rewards, the overall mission execution efficiency is lower than that in the case of sticky limit = 5. This is because when sticky limit = 7, the UAV is too greedy for target selection. As a result, there is a lack of optimization of member configuration among combat coalitions.

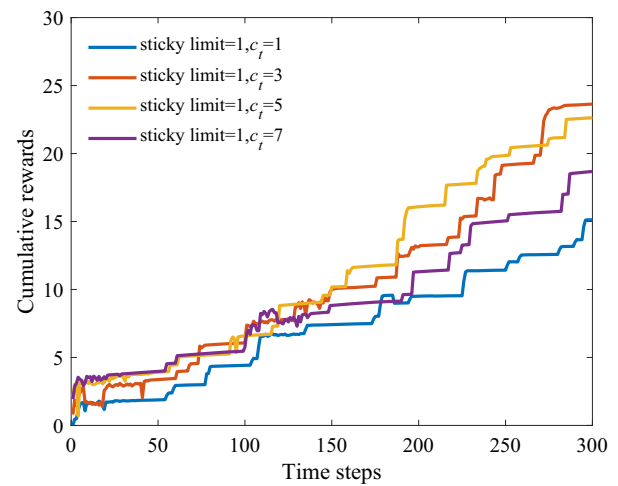
From Figs. 10, 11, 12, 13, the "confused" symptom of the swarm during the mission execution is well relieved with the increase in the sticky limit. In terms of the cumulative rewards, with the "confused" symptom well mitigated, the mission execution efficiency of the swarm is most advantageous with $c_t = 5$. At this point, the ability of the UAV swarm to reach consensus is strengthened, and the algorithm converges faster. On balance, for a mission scenario with an operational resource density of $\rho = |N_U| : |N_T| : |L^2| = 60 : 4 : 10^2$, it is optimal to set sticky limit = 5 and $c_t = 5$.

Comparison with dynamized CBGA and CBTA

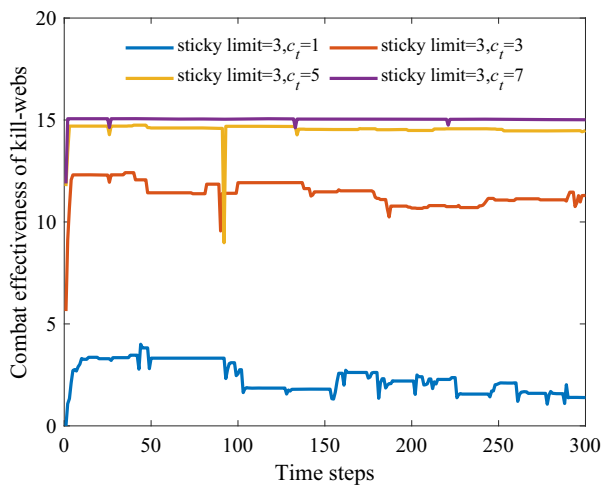
To investigate the superiority of the DCBCA algorithm in terms of solution quality and convergence speed, the CBGA algorithm [30] and the CBTA algorithm [32] are incorporated into discrete time steps while retaining the proposed dynamic improvement mechanisms, excluding the sticky limit. This leads to the DCBCGA and DCBTGA algorithms, which are then compared with the proposed DCBCA. In [36], the CBTA algorithm assists agents in fixing the target selection scheme in case of confusion by setting the maximum number of times γ that a single agent can abandon its previous target selection. If the value of γ is too small, the fixed target selection scheme



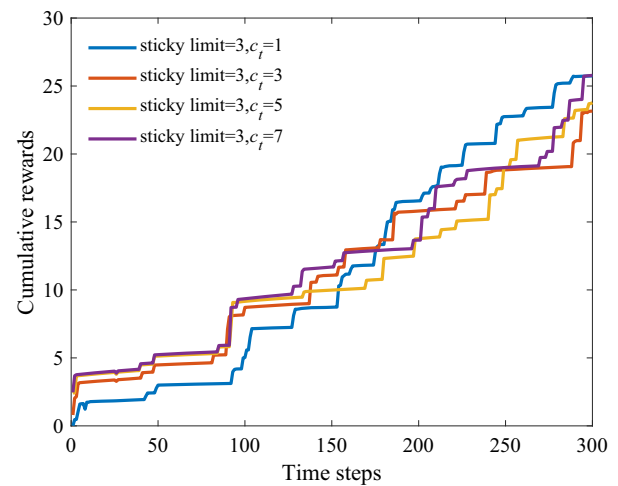
(a) Kill webs combat effectiveness.



(b) Cumulative rewards.

Fig. 10 Mission performance of the UAV swarm with the fixed sticky limit = 1

(a) Kill webs combat effectiveness.



(b) Cumulative rewards.

Fig. 11 Mission performance of the UAV swarm with the fixed sticky limit = 3

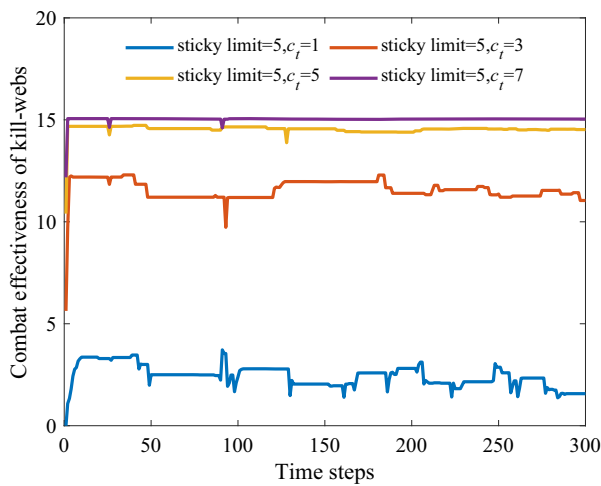
may have poor optimality, and this stubborn behavior may hinder the following consensus process; if it is too large, it loses the ability to accelerate the algorithm's convergence. Therefore, in DCBTA, γ is set to three values, 1, 3, and 5, and the corresponding algorithms are denoted as DCBTA-x. Similarly, in this paper, by setting three values of the sticky limit (3, 5, 7), DCBCA-x for performance comparison is obtained.

As proved in Sect. "Improvement of the "confused" defect", c_t is a key constraint on the performance of distributed UAV swarm missions. Therefore, for each algorithm, c_t is set to 3, 5, and 7 to examine the performance under different battlefield information conditions, remaining parameter

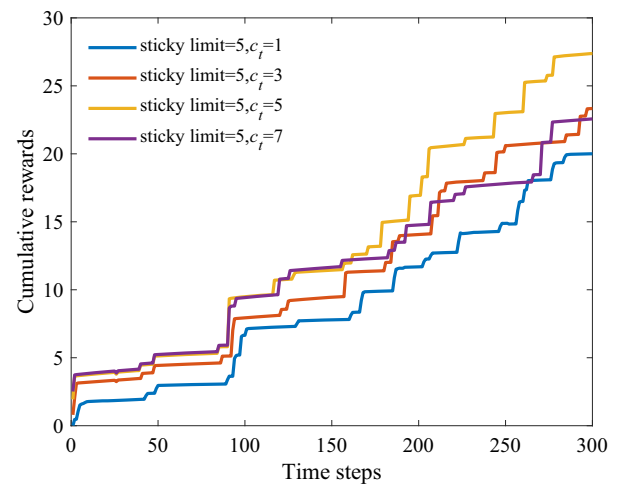
settings are consistent with those in Sect. "Improvement in the "forgetful" defect".

In Fig. 14, a comprehensive comparison of the convergence speed and solution quality of different algorithms is presented using cumulative rewards curves. The solid line represents the average cumulative rewards of the algorithms under varying information conditions, while the shaded area represents half of the standard deviation of the results in the three scenarios.

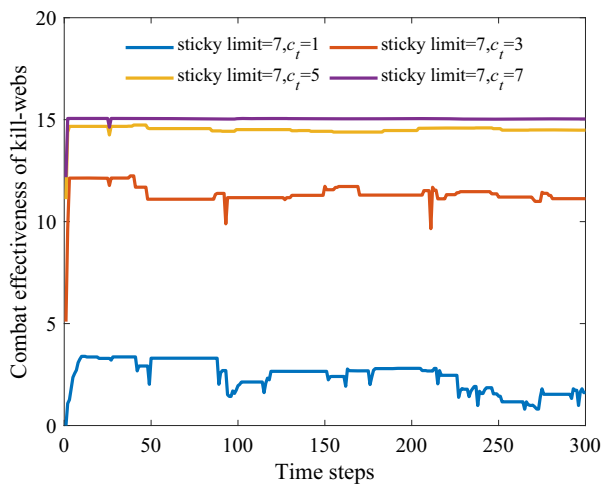
It is evident that the proposed DCBCA algorithm consistently outperforms the other two algorithms, irrespective of the sticky limit. Specifically, when γ is appropriate, DCBTA-3 outperforms DCBGA; conversely, when γ is too small, the



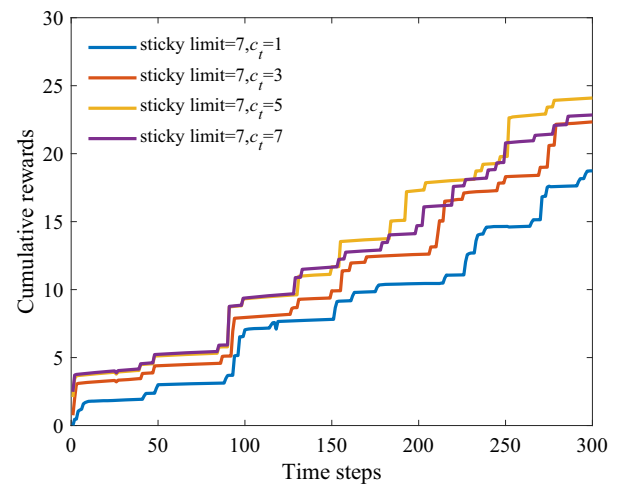
(a) Kill webs combat effectiveness.



(b) Cumulative rewards.

Fig. 12 Mission performance of the UAV swarm with the fixed sticky limit = 5

(a) Kill webs combat effectiveness.



(b) Cumulative rewards.

Fig. 13 Mission performance of the UAV swarm with the fixed sticky limit = 7

final reward of DCBTA-1 is even lower than that of DCBGA. Furthermore, in contrast to DCBTA, the proposed DCBCA algorithm inherently incorporates the geographical criteria through the setting of the sticky limit, rendering the early fixed selection scheme more reasonable and consequently facilitating a smoother consensus process.

Figure 9 reveals that as the "forgetful" defect improved, the number of trajectory planning instances for UAV swarms when dealing with pop-up targets serves as an intuitive indicator of the algorithm's convergence speed. Specifically, when this indicator reaches 0, the algorithm converges. In light of this, Fig. 15 demonstrates that the convergence speed of the proposed DCBCA is notably faster than that of the DCBGA and DCBTA. Notably, both DCBTA-1 and

DCBCA-7 exhibit the poorest performance among the same group of algorithms, because of their excessively greedy characteristics.

Resilience analysis for the UAV swarm

In this section, the proposed algorithm system is examined in contested scenarios. Assume that an external attack occurs at the 50th time step, the impact of the attack is simulated by randomly removing a specific number of UAVs from the swarm. Following this, the information about the damaged UAVs is widely disseminated, and the UAV swarm commences kill webs recovery after 5 time steps. As an illustration, a visual representation of the resilience process of the UAV swarm

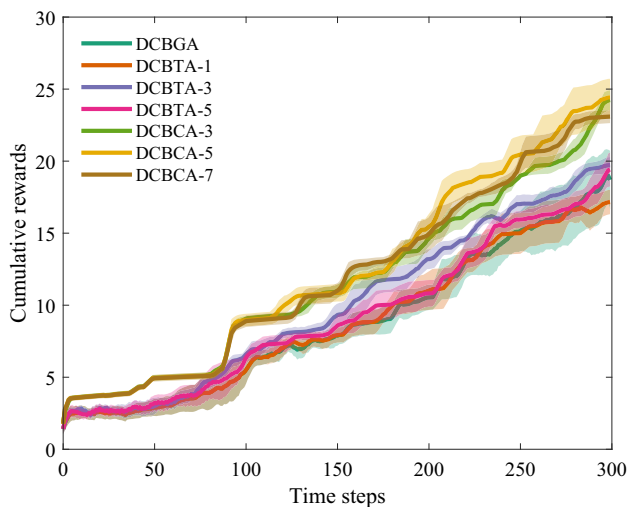


Fig. 14 Cumulative rewards of different algorithms

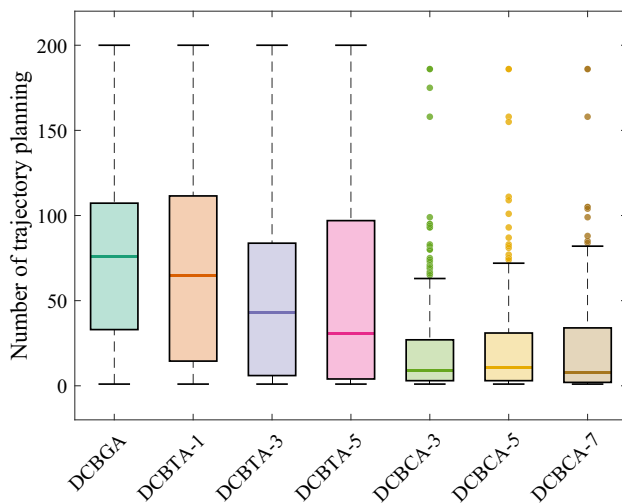


Fig. 15 Comparison of the convergence speeds of different algorithms

using an attack intensity of 10 UAVs is presented. To build a hierarchical attack intensity while avoiding the premature collapse of the swarm, the operational resource density in the mission scenario for this experiment is appropriately relaxed. Specifically, the UAV swarm size is set as $N_U = 120$, and the number of targets is set as $N_T = 4$. The rest of the parameter settings remain the same as in Sect. "Construction of kill webs with heterogeneous UAV swarms".

As shown in Fig. 16a, the UAV swarm achieves a complete kill webs construction for four targets in the mission scenario before being attacked. At the 50-time step, the UAV swarm suffers a random external attack with a total loss of 10 UAVs, which are marked with red hexagonal stars in Fig. 16b. Consequently, several gaps emerge in the kill webs against the four targets. The UAV swarm starts to recover the kill webs

after 5 time steps and just one time step later, the UAV swarm completes the deconflction of the target selection scheme.

To further investigate the mission execution capability of heterogeneous UAV swarms under contested environments, a resilience analysis [11] is developed with the real-time mission performance as the metric. By integrating the immediate and enduring combat capabilities of UAV swarms, the mission performance is defined as $F(t) = E_G(t) + \omega P(t)$, where "real-time combat effectiveness" E_G (Eq. (15)) is amalgamated with "potential combat effectiveness" P (Eq. (22)) and ω is the weight coefficient.

The resilience process of a UAV swarm combat system consists of two main parts: the absorption of disruptions and the recovery of its combat capability. For the former, random attacks [40] on different numbers of UAVs are conducted to simulate external attacks with different intensities. In the recovery phase, the proposed network adaptive mechanism and the DCBCA are utilized to recover the mission performance. The real-time mission performance under various attack intensities is presented in Fig. 17.

The UAV swarm achieves effective resilience recovery and maintains relatively stable mission performance when confronted with attacks of varying intensities, thereby highlighting the applicability of our algorithm framework in dynamic, contested scenarios.

Conclusions and future work

This study focuses on facilitating combat-oriented heterogeneous UAV swarms to construct kill webs in dynamic contested environments. The main contributions of this study, encompassing the confluence of communication networking, self-organized movement, and information decisions, are as follows:

1. A dynamic adaptive communication network model incorporating highly dynamic characteristics and communication range limitations is designed to improve the robustness of the UAV swarm communication network.
2. Based on the ABM approach, the heterogeneous UAV swarm is modeled as a multilayer networked system, which supports its self-organized operation in dynamic combat scenarios.
3. The DCBCA algorithm is proposed to solve the kill webs construction problem. The simulation results show that following the systematic improvement of multiple intelligent defects of agents, the proposed algorithm exhibits superior adaptability, and the combat system demonstrates good resilience.

For future work, the following research is planned.

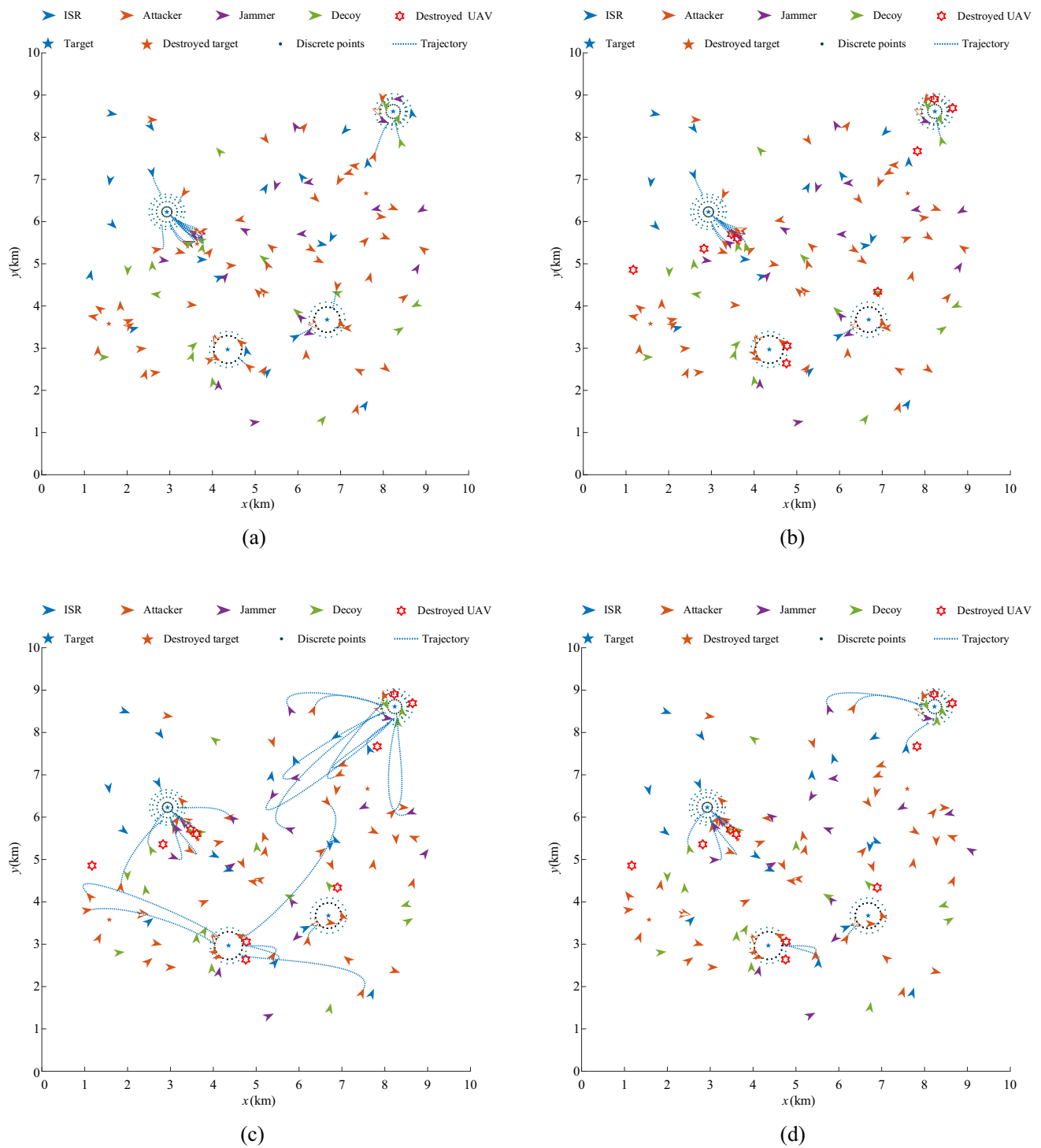


Fig. 16 Resilience process at different time steps: (a) 49, (b) 50, (c) 55, (d) 56

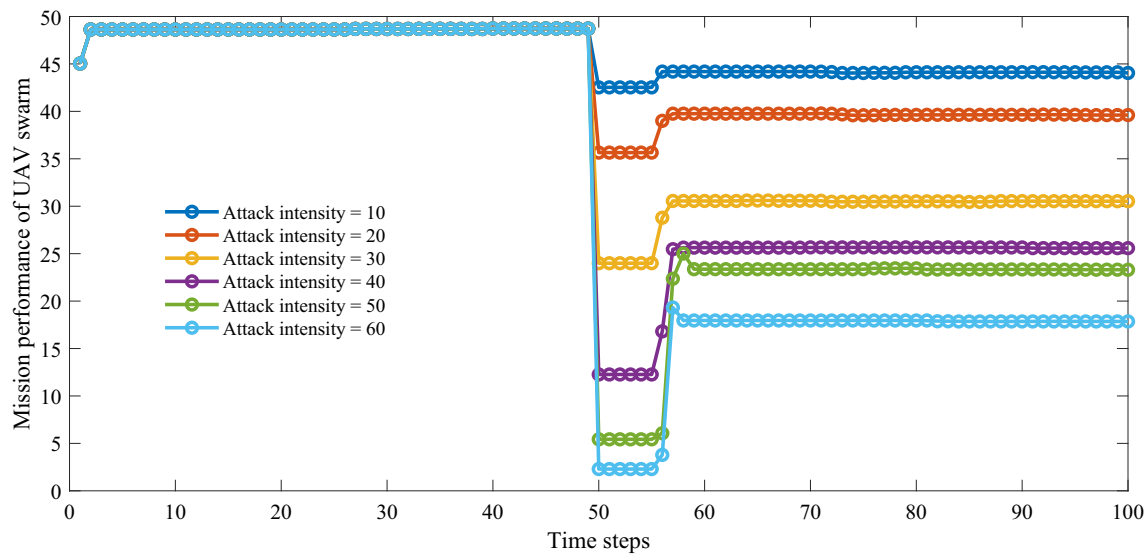


Fig. 17 Mission performance of the UAV swarm in different contested scenarios

1. The networked modeling of heterogeneous combat systems will be optimized by considering the heterogeneous properties of the combat nodes [41].
2. Self-organized behavior-driven mechanisms will be optimized by adopting more advanced motion planning/control methods [42, 43].
3. The learning evolution capability of combat systems will be enhanced by employing artificial intelligence-based approaches [44].

Author contributions Wenlin LIU: conceptualization, formal analysis. Zishuang PAN: methodology, writing—original draft. Wei HAN: investigation. Xichao SU: writing—review and editing. Dazhao YU: resources, supervision. Bing WAN: data curation, software.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. On behalf of all authors, the corresponding author states that there is no conflict of interest.

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